Chapter 20 worked solutions

Search, Retrieval, and Grounded Assistants

Exercises 1–4 work through retrieval scores, passage packing, and evidence recall by hand. Exercises 9–11 design checks for passage boundaries, abstention, and notice versions.

1. Compare two query words

Imagine searching ten office notices. Two query words occur equally often in the notice we are scoring. Across the whole collection, one word appears in only two notices and the other appears in eight. A match on the rarer word helps distinguish this notice from more of the collection.

BM25 captures that difference with inverse document frequency (IDF). The document frequency $\mathrm{df}(w)$ counts how many passages contain the word w. With ten passages, the two IDFs are

$$\begin{aligned} \text{IDF (rarer word)} &= \ln\left(1 + \frac{10 - 2 + 0.5}{2 + 0.5}\right) = \ln(4.4) \approx 1.482, \\ \text{IDF (more common word)} &= \ln\left(1 + \frac{10 - 8 + 0.5}{8 + 0.5}\right) = \ln\left(\frac{22}{17}\right) \approx 0.258. \end{aligned}$$

For a candidate containing both words with equal frequencies and length adjustments, BM25 multiplies each IDF by the same positive factor. The word found in two passages contributes more because its IDF is larger.

2. Change the query direction

The figure's original query points along the first axis. Changing the query to $(0, 1)$ points it along the second axis. The dot product multiplies corresponding coordinates and adds the results, so the first coordinate of each passage now contributes zero.

Passage	Passage vector	Dot product with $(0, 1)$
P1	$(0.8, 0.6)$	$0(0.8) + 1(0.6) = 0.6$
P2	$(0.6, 0.8)$	$0(0.6) + 1(0.8) = 0.8$
P3	$(0, 1)$	$0(0) + 1(1) = 1.0$

The ranking from highest to lowest is **P3, P2, P1**.

For a query (x, y) with $x, y > 0$, P1 has a first-coordinate advantage of 0.2 and a second-coordinate disadvantage of the same size. Subtracting the score of P2 from the score of P1 gives

$$(0.8x + 0.6y) - (0.6x + 0.8y) = 0.2(x - y).$$

P1 has the higher score exactly when $x > y$, since the query gives more weight to the first coordinate. If $x = y$, the scores tie and the chapter's rule for breaking ties by ID places P1 first.

3. Fit a short context

With 11 tokens available, P1 fits and uses nine. The rule visits every passage in rank order. Whenever a passage is too long for the space left, it skips that passage and checks the next one.

Passage	Cost	Tokens before	Decision	Tokens after
P1	9	11	Keep	2
P2	5	2	Skip	2
P3	4	2	Skip	2
P4	2	2	Keep	0

The packed context is **P1, P4**. P4 still fits after the two higher-ranked passages are skipped.

A budget of **14 tokens** keeps P2 and skips P3. P1 goes first and leaves $14 - 9 = 5$ tokens. P2 uses all five, so P3 and P4 are skipped. Ranking gives P2 the first chance at those five tokens, even though P3 is shorter. The passage lengths determine which candidates can use the space that remains at their turn.

4. Count missing evidence

The question needs two passages, P2 and P4. The first search result is P3, so we have no evidence after reading one result. The second result is P4, which supplies one of the two required passages. P2 arrives fourth.

Recall at rank k is the fraction of all required passages found among the first k results. The denominator stays at two at each cutoff. The two yes-or-no checks ask whether the search found at least one required passage and whether it found the complete set.

Rank k	Required passages found	Recall at k	Any evidence?	Both required?
1	None	$0/2 = 0$	No	No
2	P4	$1/2 = 0.5$	Yes	No
4	P4, P2	$2/2 = 1$	Yes	Yes

At rank two, an any-evidence check reports a hit while recall shows that half the required evidence is still missing.

9. Choose a passage boundary (optional)

Suppose the office posts a two-sentence notice. *Renew a parking permit online. If your address changed this month, renew in person at the mobility desk.* A resident asks, *Can I renew online if I moved this month?* The first sentence gives the ordinary route. The second sentence overrides it for this resident. To answer a question about the general rule and its exception, the system must preserve both sentences and the condition connecting them.

Build two indexes over the same notices. Use this notice in place of the chapter's P1, keeping P2 and P3 as distractors. In the first index, store the two sentences separately. Give them distinct passage IDs and retain the document ID and each sentence's location. In the second, index the whole two-sentence notice under one passage ID. Keep the surrounding notices and their passage boundaries fixed. In the split index, a query that matches words in the first sentence can retrieve that sentence while leaving the exception behind. In the joint index, retrieving and packing the entire notice brings both facts into the prompt. A combined passage also costs more tokens than either sentence alone, so a small budget may exclude it. Splitting can therefore help fit the one sentence a particular question needs.

Test all three ways of asking about the notice. Ask about the ordinary route, the exception for a recent move, and both routes in one answer. For the last case, ask *Where should residents renew, and how does the procedure change if they moved this month?* Add paraphrases such as *I relocated this month. How do I extend my parking authorization?* Before running retrieval, label the original sentence spans that support each answer. Use spans in the original notice as the scoring unit in both indexes. Counting passage IDs directly would change the denominator when two passages become one.

Use the same queries, retriever, rank cutoff, packing policy, token budget, and generation settings. Rebuild the index for each boundary scheme. Record whether each required sentence appears in the top candidates and whether it actually reaches the final input. Report coverage of the exception and, for questions requiring both parts, the share with both spans available. Run a second, larger budget to see which differences come from fitting the passages in the prompt. For example, if the split index finds the exception at rank two but the prompt contains only the first sentence, retrieval found the evidence and packing discarded it.

Score the answer separately. For the resident who moved this month, credit an answer that directs the resident to renew in person at the mobility desk and ties that instruction to the recent address change. For the question asking about both routes, require the default online route and the conditional in-person exception. Check the IDs against the passages actually supplied. If the exception is visible and the answer still says to renew online, the generator overlooked or misapplied available evidence. Supplying both sentences directly provides a control for whether the generator can follow the rule when search and packing have already succeeded.

10. Design an abstention test (optional)

An *abstention* is an explicit response that the available notices do not provide enough information for an answer. Use the three notices in the chapter as the entire collection. The goal is to see when the system answers and when it acknowledges missing evidence.

Question	Expected response
When does a parking permit expire?	It expires on June 30. Cite P2, which states the date.
Is there a grace period after the parking permit expires?	The notices do not specify a grace period. P2 gives the expiration date only.
Who won the district chess tournament?	The notices do not provide that information.

The middle question shares the subject and vocabulary of P2. A retriever can find P2 and a generator can quote the correct date while still failing to answer whether a grace period exists. The third question has no supporting notice either. A statement such as *There is no grace period* would go beyond the collection. The absence of a rule in these notices leaves its status unknown.

For the first question, award credit when the answer gives June 30 and cites P2 from the supplied context, with no unsupported additions such as a deadline at noon. Report factual correctness and citation support as separate checks so a correct but uncited date is visible. Count a refusal here as a missed answer in the end-to-end evaluation. For each unanswerable question, award credit when the answer clearly states that the notices are insufficient and supplies no invented fact. The response may mention a supported fact that explains the gap, such as the expiration date, as long as it explicitly withholds the requested rule. An answer that changes the subject earns no credit.

For a larger test, report the grounded answer rate on answerable questions and the appropriate abstention rate separately for the topically related and unrelated unanswerable questions. Each rate uses the number of questions in that group as its denominator. Also count refusals on answerable questions and unsupported answers on unanswerable questions. In this three-question demonstration, always abstaining would pass two cases but yield zero grounded answers in the answerable group. Report the counts as well as the rates so the small sample is clear.

Save the retrieved and packed IDs with every answer. If the expiration question receives a refusal because P2 never reached the prompt, the system missed an answer that existed in the collection. Given the actual prompt, the generator's refusal was appropriate. Compare this case with a run that supplies P2 directly. If that restores the grounded answer, check search and packing in the original run. If the system still refuses with P2 visible, examine how the generator uses the evidence. For the unanswerable questions, the gold set of supporting passages is empty, so use the abstention score instead of assigning a retrieval recall of zero.

11. Audit two versions (optional)

Keep P1 as the current renewal notice, *Renew a parking permit at the mobility desk*. Add an older passage, P1-old, that reads *Renew a parking permit at the records counter*. Give both the same office and policy identity. A query containing *renew a parking permit* matches both texts. The difference that determines the current answer is when each notice applied.

Store a distinct stable passage ID for each text version, its parent document and passage location, the issuing office or jurisdiction, and the version or revision identifier. Record the effective start and end dates, any supersession link, and the source location for citation. Keep publication and effective dates separate when they differ. In this fictional example, P1-old could apply from January 1 through March 31, 2026, and P1 from April 1, 2026, onward, with no end date yet. Use that year for the questions and save the collection snapshot. With both records retained, a citation remains traceable to the exact text the model received, and the old notice remains available for a historical question.

Create a labeled test pair with the same topic and different requested dates. For *Where do I renew a parking permit on April 15?*, require *mobility desk* with a P1 citation. For *Where were residents supposed to renew a parking permit on March 15?*, require *records counter* with a P1-old citation. Add an undated question and define its reference date in the evaluation. For example, set the test clock to April 15 and apply the notice valid on that date. Include versions of the query that use the shared phrase *renew a parking permit*, and versions that paraphrase it. Before running the system, write down the applicable ID and supported location for every case.

Compare word-only retrieval against a version-aware run under the same rank cutoff and token budget. One version-aware procedure restricts the search to notices whose office and effective interval match the question's scope and date, then ranks eligible notices. Log the candidate IDs, their dates and scores, the chosen ordering, and the final packed IDs. The two notices tie under the chapter's BM25 for *renew a parking permit*. Both have eight word tokens and contain each query word once. A ranking that happens to return P1 first because its ID sorts earlier does not establish that the system used the date. The historical query tests whether it can use the older notice when that version governs.

For each question, record whether the applicable ID is in the top results, whether it reaches the prompt, whether the answer states the applicable location, and whether the cited ID both supports that location and applies on the requested date. A current answer that says *mobility desk* but cites P1-old has the correct location and fails citation support. A current answer that cites P1-old for *records counter* faithfully quotes a passage that does not govern the question. If both notices enter the prompt, also check that the answer resolves the date rather than presenting both desks as current options. Report results for current and historical questions separately so success on one cannot hide failures on the other.